

Generation Interconnection Request System Impact Study Report

For Queue Project Q12-0181

Mitchell Dam 138 kV
January 25, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0181. The Interconnection Customer proposes a 132.6 MW natural gas combined cycle facility interconnecting at the Mitchell Dam 138 kV substation in the Virginia area. The request was received on December 02, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 139 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$799,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Lake Crk to Sierra Pacific Lumber 138 KV
Contingency: Taber Substation to Glenannan 345 KV
Loading: 103.8% (Rating: 270 MVA)
DFAX: 5.5%
Pre-project loading: 101.1%
2. Facility: Dixleta to Drum 138 KV
Contingency: Dixleta to Taber Substation 345 KV
Loading: 102.1% (Rating: 234 MVA)
DFAX: 5.7%
Pre-project loading: 98.9%
3. Facility: Sharyland to Tombstone 345 KV
Contingency: Tombstone to Bixby Hill 500 KV
Loading: 100.6% (Rating: 1,429 MVA)
DFAX: 7.4%
Pre-project loading: 99.9%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Sierra Pacific Lumber to Lake Crk 138 KV Line	Reconductor line	\$864,000	4% (\$37,000)
Dixleta to Drum 138 KV	Reconductor line	\$1,081,000	4% (\$43,000)
Tombstone Substation to Sharyland 345 KV	Reconductor line	\$25,619,000	3% (\$719,000)
	Total allocated to project		\$799,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.